Paper 001 · Institutional Design for an AI Society

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Question and method

How have manufacturers, users, infrastructure operators, and public institutions shared responsibility for the safety of a widely adopted technology? We study public documents on Japanese road transport, identify the functions that institutions perform, and compare them with existing AI initiatives to identify questions that arise across organizations and platforms.

Functions drawn from historical sources

Japan's National Police Agency traces the development of postwar traffic rules, the Road Traffic Act of 1960, the 1970 Traffic Safety Policies Basic Act, and subsequent measures. The Ministry of Land, Infrastructure, Transport and Tourism distinguishes registration from inspection against safety and environmental standards. We read these sources to examine how responsibilities are allocated and how institutions respond to changes during use. National Police Agency · MLIT

Road transport mechanism	Function	Question for an AI society
Registration and vehicle numbers	Identification and links to accountability	Can an operation be traced to its actor and delegator?
Licensing and administrative measures	Qualification and participation limits	Which actions can be delegated, and when can authority be revoked?
Inspection and maintenance	Continuous conformity	Does an evaluation still apply after a model or tool changes?
Traffic rules and enforcement	Shared conditions and enforcement	Can out-of-scope actions be controlled across organizations?
Road design, education, and crash analysis	Environmental safeguards, competence, and evidence feedback	Can the operating environment and human oversight be improved in light of observed outcomes?

Applying the analogy and recognizing its limits

We distinguish the safety that a developer can improve within a product from safety that several parties share in a common environment. AI already has organizational risk management initiatives and work on agent identification and authentication. A remaining research question is how to connect these to delegation, control over operations, and accountability after incidents. NIST AI RMF · NIST AI Agent Standards Initiative

Digital agents can be copied, updated, delegated to, and connected to multiple tools. A direct transfer of institutional labels would obscure these differences in identity and reach. The comparison yields hypotheses about institutional functions. It does not measure the causal effect of an individual traffic measure or the effect of transferring it to AI.

Institutional time compression

We hypothesize that capabilities, deployment, and interconnection may change faster than institutions can observe risks, build agreement, and update rules. Paper 002 considers requirements that can be tested early and revised in response to incidents and pilot evidence.

Research series · Paper 002

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